AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 22 · STUDENT LESSON

Train a Regression Model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Weeks 20–21: evaluate candidate models with a stated loss.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 19

Your mission

> Start with the target: Fit a line from a declared training candidate set. Try the recall before opening notes.

Your learning target

I can fit a line from a declared **training candidate set**.

I can separate **training** loss from validation comparison.

I can interpret **slope** and **intercept** with units.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.



Your target

LEARNING TARGET **Training** points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

Builder name and experiment date:

Readiness for the connected explanation

How is MSE calculated?

This week's end goal

By the end, I can explain train a regression model using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 19

Notice the evidence

> Investigate the supplied Train a Regression Model case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use train a regression model to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A line predicts y as m*x+b. In this classroom search we compare a small declared set of **slope**s and **intercept**s; this is not a full least-squares solver over all possible lines.

Use **training** error to choose a candidate, then freeze it before validation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **fitting** a line.

STUDENT LESSON · 3 OF 19

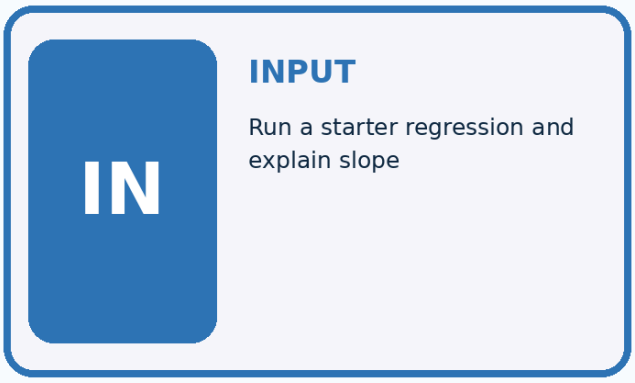
See how it works

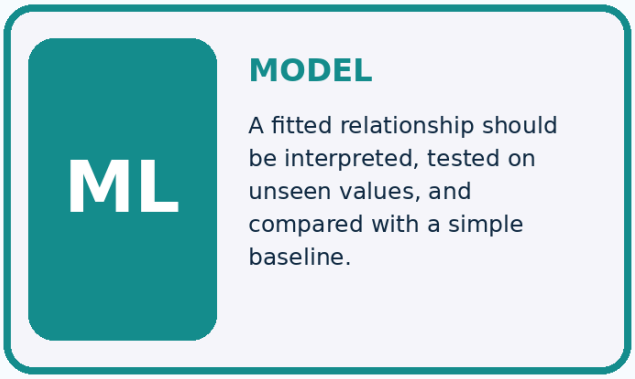
> Follow the mechanism for Train a Regression Model. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Run a starter regression and explain **slope**, **prediction**, residual, and MAE.

MODEL IDEA A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Model predicts height = 4 + 2 × weeks. **Slope** 2 means two predicted centimeters per additional week in this data range.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Training pairs are (0, 1),(1, 3),(2, 5). Candidate A has m 1 b 1, B m 2 b 0, C m 2 b 1. **Prediction**s are A [1, 2, 3], B [0, 2, 4], C [1, 3, 5]. **Training** squared-error sums are 5, 3, 0 respectively, so choose C.

Ties would use the earlier listed candidate.

```
training = [(0, 1), (1, 3), (2, 5)]
candidates = [(1, 1), (2, 0), (2, 1)]
def sse(line):
    m, b = line
    return sum((y - (m * x + b)) ** 2 for x, y in training)
chosen = min(candidates, key=sse)
print([sse(line) for line in candidates])
print(chosen)
# [5, 3, 0], then (2, 1); first candidate wins ties.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 19

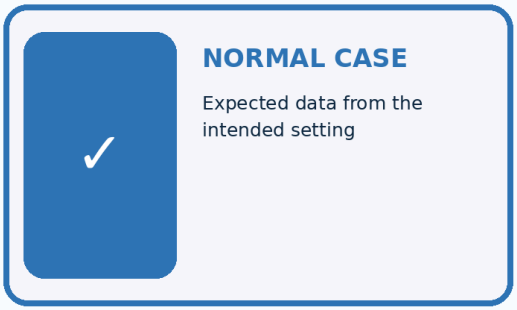
Compare the cases

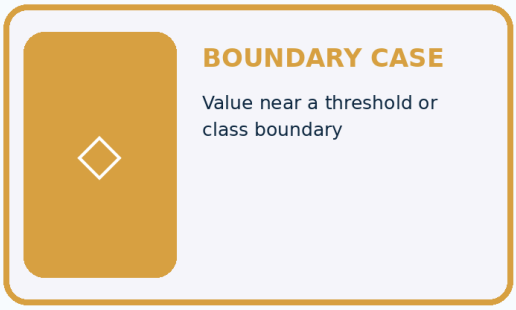
> Compare the supplied cases for Train a Regression Model. Identify the detail that changes the conclusion.

EVALUATION DESIGN

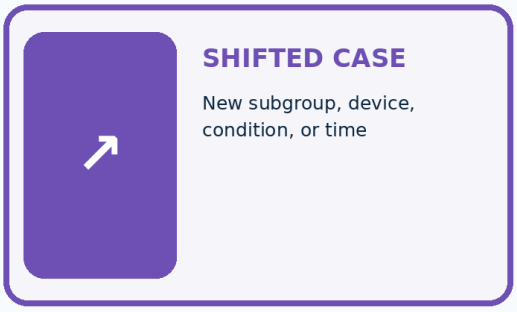
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Evaluation **observation** becomes8 instead of6.

Predicted result

Prediction: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Why the result follows

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

STUDENT LESSON · 5 OF 19

Words that unlock the idea

> Use the vocabulary of Train a Regression Model to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

COEFFICIENT

learned relationship weight

2

INTERCEPT

prediction when inputs are zero

3

SLOPE

predicted change per input unit

4

GENERALIZATION

performance on unseen cases

Term	Precise meaning	My ML example
coefficient	learned relationship weight	
intercept	prediction when inputs are zero	
slope	predicted change per input unit	
generalization	performance on unseen cases	

MISCONCEPTION TO REPAIR A positive **slope** proves one variable causes the other.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Slope	Change in predicted target per unit change in x
Intercept	Predicted target when x is zero
Candidate set	The declared model choices searched
Fitting	Selecting model values using training evidence

STUDENT LESSON · 6 OF 19

Follow a worked example

> Follow the worked trace for Train a Regression Model, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 22 • WORKED EVIDENCE

1. Model predicts height = 4 + 2 × weeks.
2. Slope 2 means two predicted centimeters per additional week in this data range.
3. At week 5, prediction is 14 cm; if actual is 16, residual is +2 cm.
4. Evaluate unseen plants and compare test MAE with the median baseline.

Step 1

STEP 1 Model predicts height = 4 + 2 × weeks.

Step 2

STEP 2 **Slope** 2 means two predicted centimeters per additional week in this data range.

Step 3

STEP 3 At week 5, **prediction** is 14 cm; if actual is 16, residual is +2 cm.

Step 4

STEP 4 Evaluate unseen plants and compare test MAE with the median baseline.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Fitting a line at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Predict at x 3 and x 4 and calculate MAE/MSE. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train a Regression Model

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

The facts we will use

Supplied facts

Training(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/22

Train a Regression Model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Calculate training fit

2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5.

2. Select line

Choose2x+1 under minimum training SSE.

3. Evaluate once

At day3 predict7 versus observed6: absolute error1 cm.

CHANGE ONE THING

Evaluation observation becomes8 instead of6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Calculate training fit	2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5.
2. Select line	Choose2x+1 under minimum training SSE.
3. Evaluate once	At day3 predict7 versus observed6: absolute error1 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Keep the stated limit

Read the labelled picture: May day3’s evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Evaluation **observation** becomes8 instead of6. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 19

Find and repair a mistake

> Find the unsupported step in the Train a Regression Model claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A positive slope proves one variable causes the other. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A positive **slope** proves one variable causes the other.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Record candidates, **training** rows, tie policy and chosen parameters. Do not add a candidate after seeing validation and report it as if only the original search happened.

A zero **training** loss can coexist with validation mistakes.

Why is an exact training line not proof of correct predictions at x 100?

STUDENT LESSON · 8 OF 19

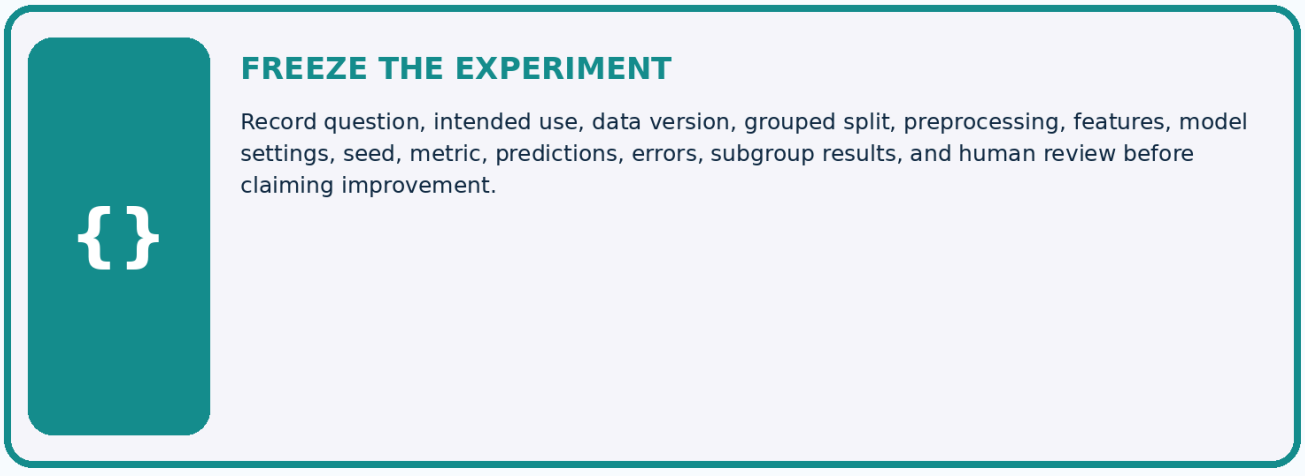
Practise together

> Practise Train a Regression Model with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If x measures days and y measures cm, **slope** 2 means two predicted cm per day and **intercept** 1 means one predicted cm at day zero.

Predicting far outside the observed x range is extrapolation and may fail even when the **training** fit is exact.

Practise with less help: Score validation

Use workbook W2 for the connected example "**Fitting** a line". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Fitting** a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

May day3’s evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 19

Run the investigation

> Investigate Train a Regression Model. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Regression Run: trace code, interpret coefficients with units, predict unseen values, and compare MAE with baseline.

1. Write the hypothesis and expected **evidence** before running.

2. Regression Run: trace code, interpret **coefficient**s with units, predict unseen values, and compare MAE with baseline.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

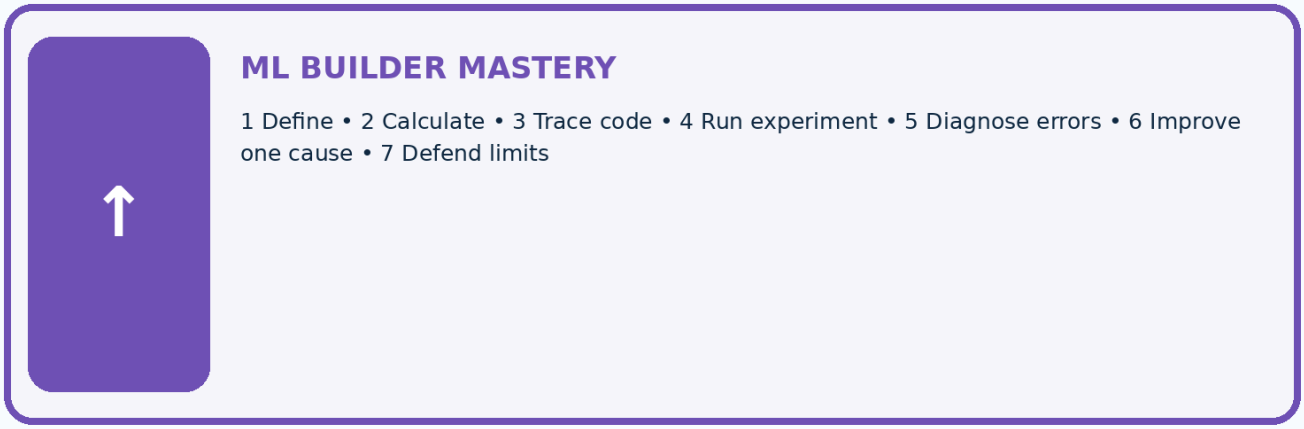
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Train a Regression Model. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **coefficient**, **intercept**, **slope**, **generalization**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A positive **slope** proves one variable causes the other.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 19

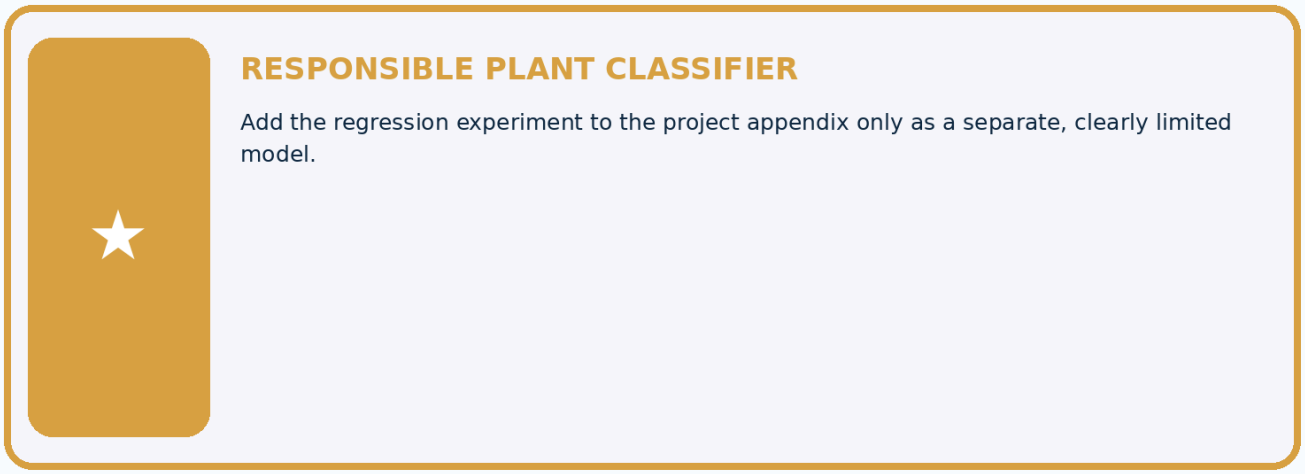
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Add the regression experiment to the project appendix only as a separate, clearly limited model.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 19

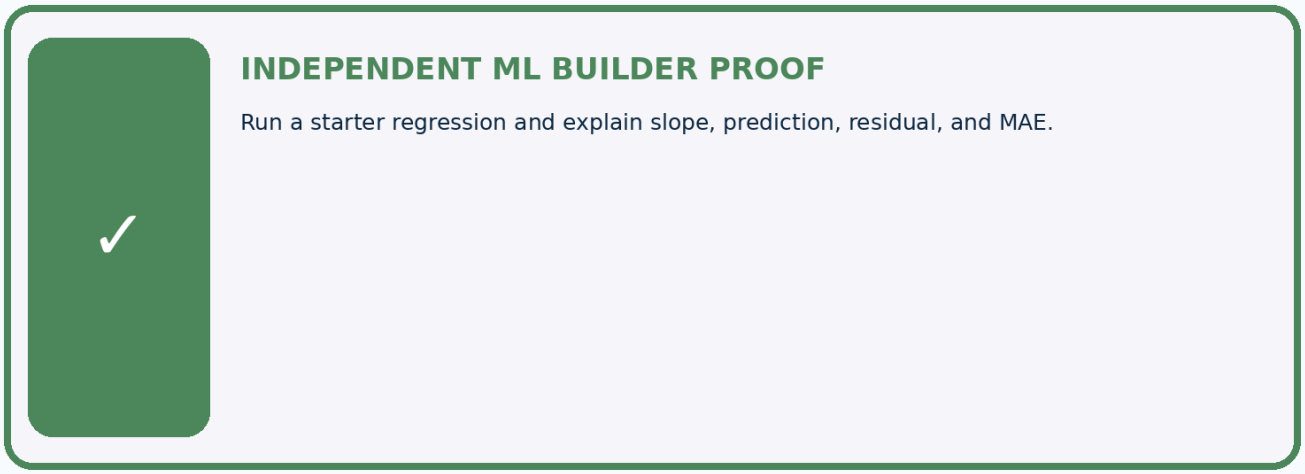
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Run a starter regression and explain **slope**, **prediction**, residual, and MAE.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Hold Out Test Data.

Study a complete case

> Revisit the complete Train a Regression Model case for supported practice. Repeated examples do not become fresh assessments.

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

The question

Predict and evaluate; distinguish inference.

Worked explanation

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Apply the fitted equation, then evaluate

Evidence or stage	Value or result
Rule	height = 2 × age + 1
Age 3	Predict 7, actual 8, absolute error 1
Age 4	Predict 9, actual 9, absolute error 0
MAE	0.5 cm

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

STUDENT LESSON · 14 OF 19

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10), (4, 15) with MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A regression model has validation MAE 2.4 cm; a **training**-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add the regression experiment to the project appendix only as a separate, clearly limited model.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Training points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 19

Fit and test a straight-line regression model

> Follow the supplied Train a Regression Model lab. Predict first, keep actual results and explain the changed test.

Estimate **slope** and **intercept** from **training** pairs using least squares. Ages are weeks and heights are cm. Test targets are excluded from **fitting**; **training** ages must include distinct values.

Math bridge: fitting the line

A straight line predicts y = **slope** × x + **intercept**. **Slope** is the change in predicted height for one extra week; **intercept** is the **prediction** at x=0, which may lie outside the measured ages.

You need means, subtraction, multiplication and squared numbers for this example; calculus is not required.

x (weeks) · y (cm) · x – mean x · y – mean y · Product · Squared x difference

> **x (weeks):** 1
> **y (cm):** 3
> **x – mean x:** –1
> **y – mean y:** –2
> **Product:** 2
> **Squared x difference:** 1

> **x (weeks):** 2
> **y (cm):** 5
> **x – mean x:** 0
> **y – mean y:** 0

> **Product:** 0
>
> **Squared x difference:** 0

> **x (weeks):** 3
>
> **y (cm):** 7
>
> **x – mean x:** 1
>
> **y – mean y:** 2
>
> **Product:** 2
>
> **Squared x difference:** 1

Mean x=2 and mean y=5. Add the product column: 4. Add squared x differences: 2. **Slope**=4/2=2 cm per week. **Intercept**=5−2×2=1 cm.

This least-squares formula chooses the line minimising squared **training** errors when the x values are not all identical. Compare its held-out error with a baseline; a good **training** fit alone is insufficient.

```
"""Fit a line to training pairs and evaluate a separate example."""
train = [(1, 3), (2, 5), (3, 7)]
if not train:
    raise ValueError("Training examples are required")
n = len(train)
x_mean = sum(x for x, y in train) / n
y_mean = sum(y for x, y in train) / n
num = sum((x - x_mean) * (y - y_mean) for x, y in train)
den = sum((x - x_mean) ** 2 for x, y in train)
if den == 0:
    raise ValueError("Distinct training ages are required")
slope = num / den
intercept = y_mean - slope * x_mean
test = [(4, 10)]
if not test:
    raise ValueError("Test examples are required for MAE")
preds = [slope * x + intercept for x, y in test]
mae = sum(abs(y - pred) for pred, (x, y) in zip(preds, test)) / len(test)
print(slope, intercept, preds, mae)
```

Expected **output**: 2.0 1.0 [9.0] 1.0

The fitted line is height = 2 times age + 1. At four weeks it predicts 9 cm; the observed height of 10 cm gives an absolute error and one-case MAE of 1 cm. **Fitting** a line and applying a supplied line are different activities.

One held-out case is very limited **evidence**.

Independent code change

Change the held-out height from 10 to 8 without changing the **training** data. Predict the fit, **prediction** and MAE.

My **prediction** and **evidence**

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 16 OF 19

Your lesson route

> Use this map to navigate Train a Regression Model, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

By the end, I can	How I will show it
Fit a line from a declared training candidate set	Explain the deciding step, show a trace or test, and state one limit.
Separate training loss from validation comparison	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Interpret slope and intercept with units	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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Finish and return

> Explain Train a Regression Model without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Fit a line from a declared **training candidate set**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At day3 predict7 versus observed6: absolute error1 cm.

1. Select line

2. Calculate **training** fit

3. Evaluate once

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage

that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Evaluate once At day3 predict7 versus observed6: absolute error1 cm. Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Evaluation **observation** becomes8 instead of6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Absolute evaluation error stays1 cm, but residual changes from-1 to+1. **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Repair a missing building block

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

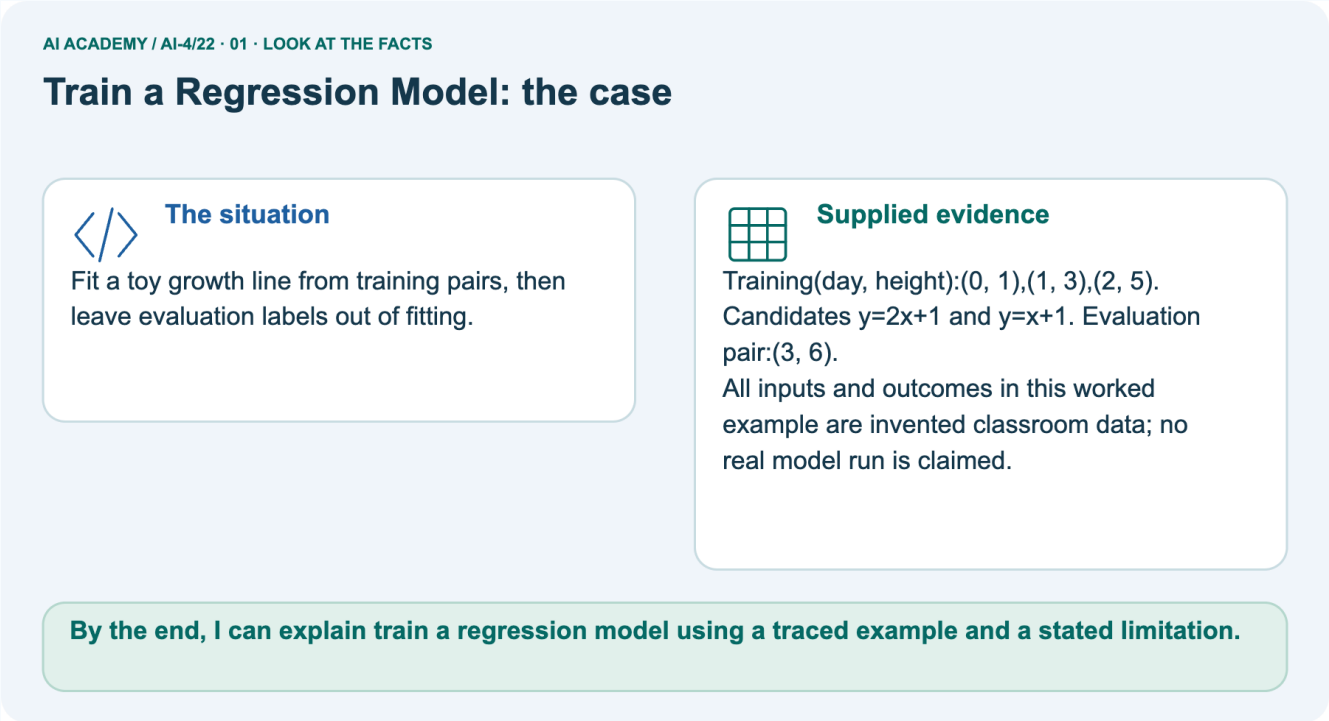
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Train a Regression Model graphic using its labels, deciding evidence and relevant limitation.

01. Train a Regression Model: the case



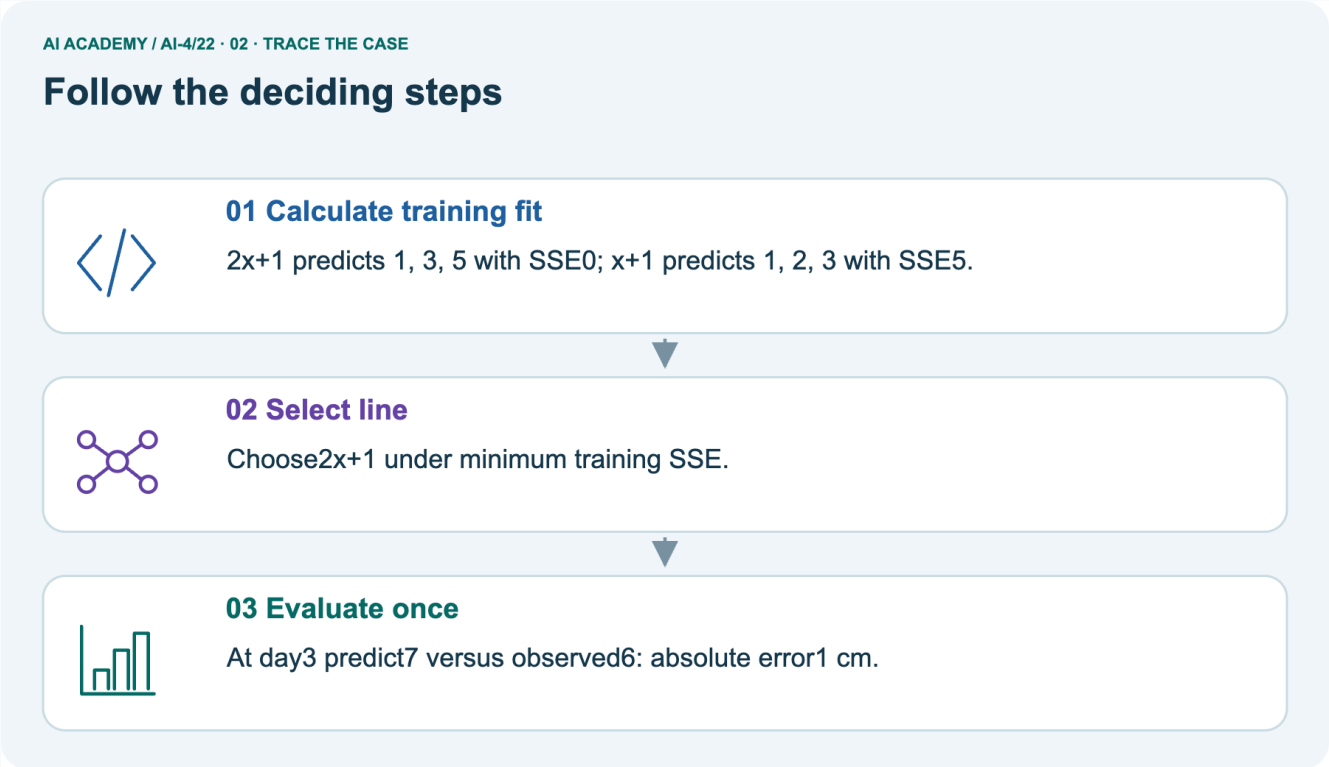
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**. Supplied facts: **Training**(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **training** fit: 2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5. → Select line: Choose2x+1 under minimum **training** SSE. → Evaluate once: At day3 predict7 versus observed6: absolute error1 cm.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Fitting a line at a glance

AI ACADEMY / AI-4/22 · 03 · INSPECT THE LESSON MODEL

Fitting a line at a glance

Part	Evidence or action	Interpretation
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

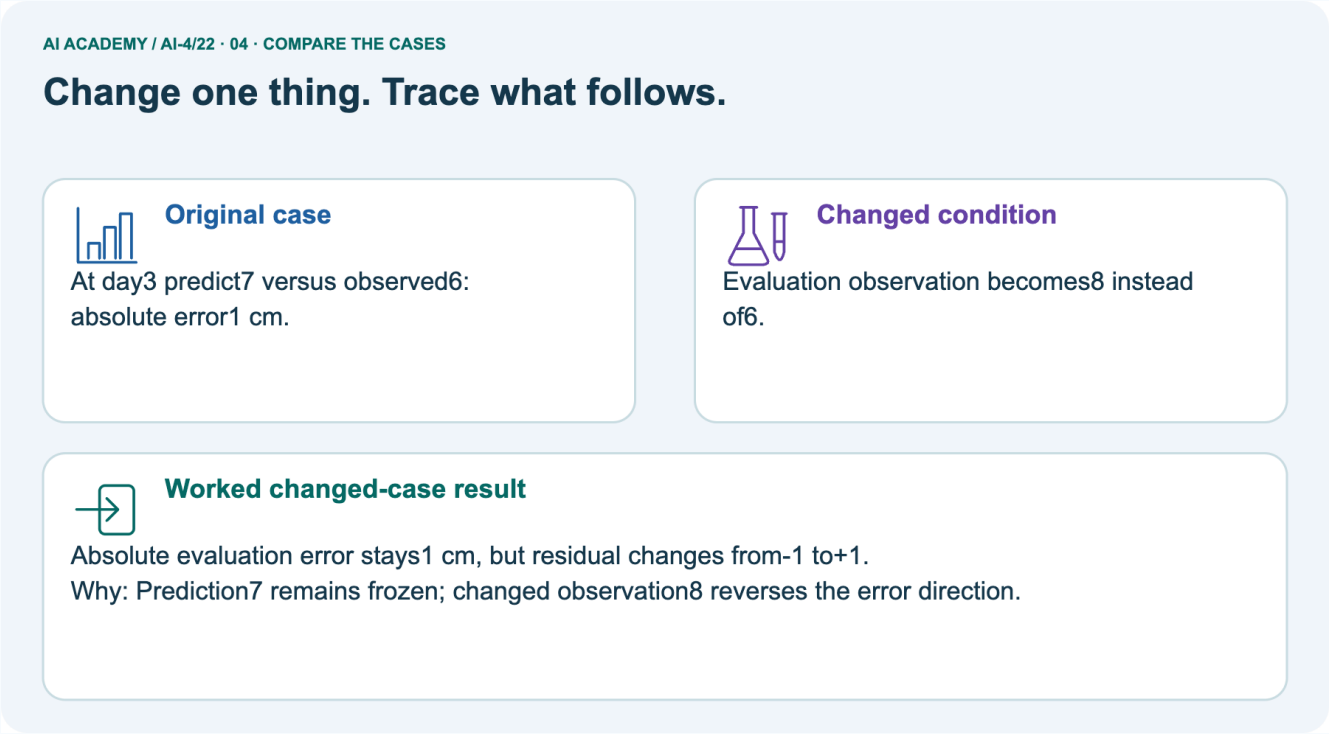
For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Follow the labelled picture with the matching explanation. Open the vector diagram

Fitting a line at a glance. Line A | **Slope** 1 **intercept** 1 | **Training** SSE 5. Line B | **Slope** 2 **intercept** 0 | **Training** SSE 3. Line C | **Slope** 2 **intercept** 1 | **Training** SSE 0

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



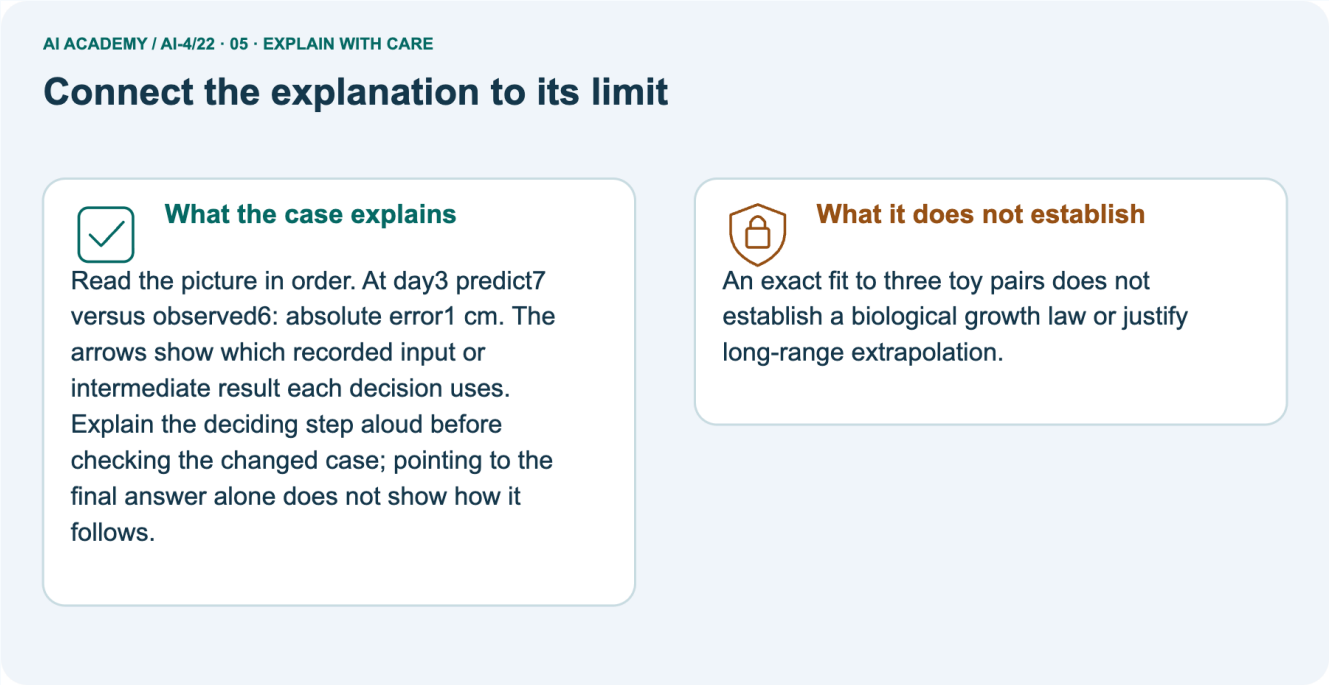
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At day3 predict7 versus observed6: absolute error1 cm. Changed condition: Evaluation **observation** becomes8 instead of6. Expected result: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

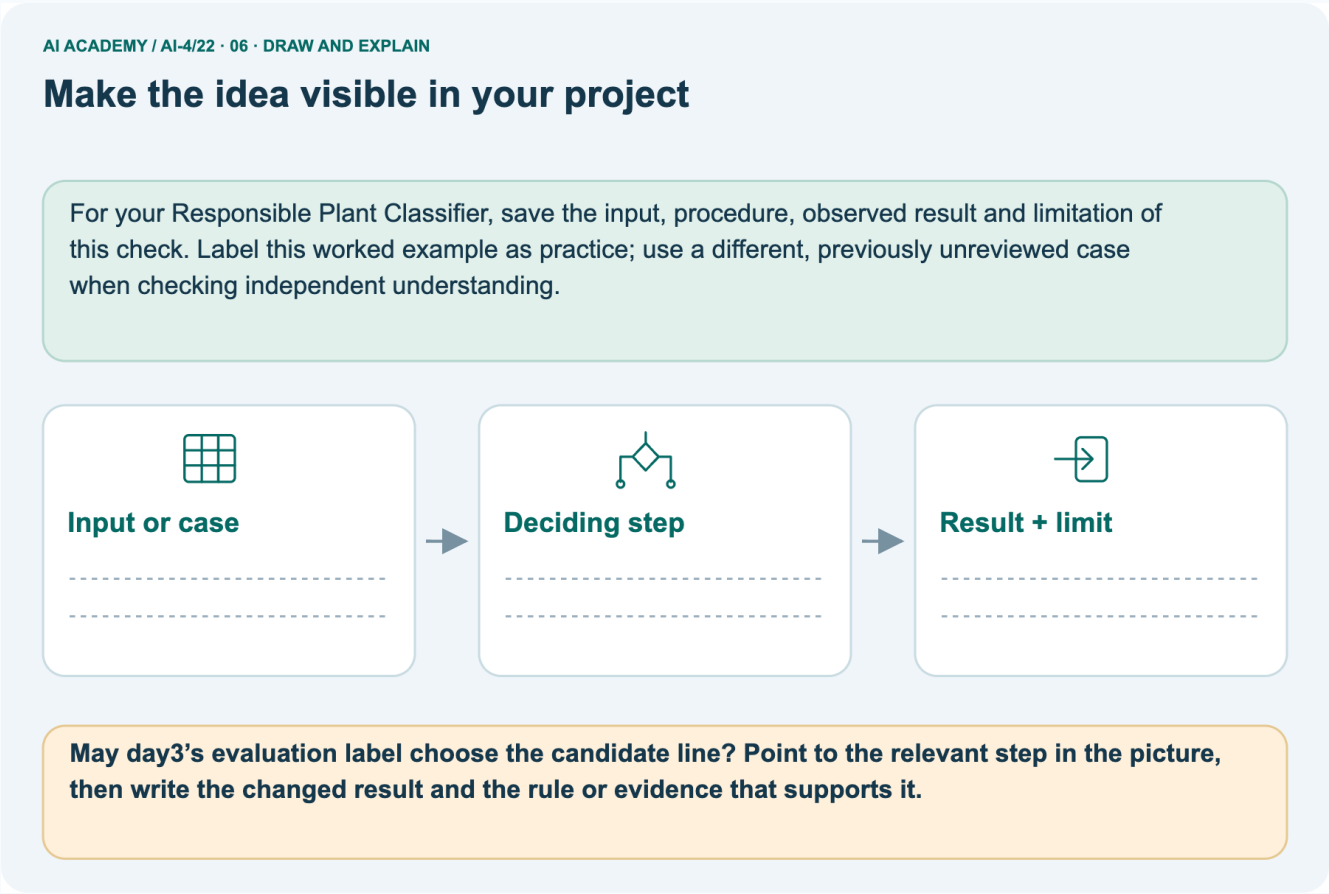
Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.